Mike Ndlovu

Warsaw Poland   mikendlovu771@gmail.com   +48 515 954 304

mike-ndlovu-bbb843251/   Gabbzgreat

Summary

Electrical Engineering student passionate about aerospace, embedded systems, and intelligent control software. Experienced in MATLAB/Simulink modeling, PLC logic design, and Python programming. Skilled in developing reliable, safety-critical, and data-driven systems. Eager to contribute to robotics and automation innovation at academic and industrial level through model-based design and software engineering excellence.

Education

BSc **Warsaw University of Technology**, Electrical Engineering — Oct 2022 – Mar 2026

- Rector's Scholarship of Academic Excellence (2023–2024)
- Focus on control systems, hardware diagnostics, and embedded solutions

Experience

Triol Corporation **Electrical & Automation Engineering Intern** — Ostrówek, Poland — Jul 2024 – Sep 2024

- Modeled and validated AC–DC–AC converters in **MATLAB Simulink** for VFD control verification
- Developed V/f regulation logic in SCL and MATLAB, improving control precision by 15%
- Automated data collection and diagnostic scripts, reducing manual troubleshooting by 30%
- Collaborated with R&D to verify embedded control software and hardware integration

Projects

Centrifuge Control System (TIA Portal – SCL)

- Designed a PLC-based control system with seven operational states, fault handling, and restart safety logic
- Simulated process control behavior in **Simulink** to validate timing and sensor response
- Created an HMI interface (TP700 Comfort) for process visualization and diagnostics

PCB Component Detection using YOLOv8

- Developed a computer vision tool to classify and verify PCB layouts using **YOLOv8**
- Achieved 89.2% precision and integrated a Python-based UI for automated inspection

Fuse & Wire Calculator (Python GUI)

- Built a Python GUI tool to automate voltage/current calculations and improve lab workflow efficiency by 35%
- Applied algorithmic optimization and data structures for performance improvements

Skills

Programming & Modeling: C, Python, MATLAB, Simulink, Siemens SCL

Control & Automation: PLCs (TIA Portal), HMI Panels, LabVIEW, Control Logic Design

Diagnostics & Testing: AC–DC conversion simulation, fault diagnosis, embedded safety logic

Hardware & Tools: Circuit testing, Altium Designer, LTspice, Git, Excel

Languages: English (C1), Polish (A1 – learning)